



# Extraction, Mapping, and Analysis of Structural Lineaments using Geospatial Azimuth-Altitude Ratio Technique on a DEM of a Sub-Catchment of the Upper Benue River Basin, Nigeria

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## Abstract:

Lineaments have a wide range of implications and applications in geology, hydrogeology and geomorphology. The focus of this study was on structural or geological lineaments and their implication on hydrogeology and hydrogeomorphology. Hence, this study aimed at the extraction, mapping, and analysis of structural lineaments using a DEM in the Adamawa sub-catchment of the Upper Benue River basin. ArcGIS 10.8 was used for the geospatial technique of Azimuth-Altitude ratio (315:45, 200:50, 150:65, and 100:60) variation, manual digitization for the extraction, mapping and analysis of lineaments. The choice of manual or visual digitization was to apply expert experience to extract only structural and exclude

geomorphological and pseudo lineaments. The Azimuth-Altitude ratio variation approach was to improve clarity and identification to aid the manual extraction. The lineament map generated after extraction was used to analyze for lineament density, total number, and total length. A further analysis applied the rose diagram to depict the directional orientation or trends of these lineaments. The result revealed that at various Azimuth and Altitude ratios, the features on the image become more distinguishable for extraction. At 315:45 Azimuth and Altitude ratio, the hill-shade produced revealed the major lineaments more clearly compared to other ratios but minor lineaments were undermined. Also, a total number of 343 lineaments were found in the Adamawa sub-catchment of the Upper Benue River Basin with a total length of 2897.81km. The lineament density was found to range between 1km/km<sup>2</sup> and 5km/km<sup>2</sup>. The result of the lineament statistical analysis showed that the extracted lineaments have trends in the direction of N-S, NNE-SSW, NE-SW, E-W, SE-NW, and SSE-NNW. This therefore concluded that the Adamawa sub-catchment of the Upper Benue River Basin has high lineament density and lineament extraction from DEM at various Azimuth and Altitude ratios enhances lineament extraction. The implications of the findings from the results suggest that areas with a high density of lineaments might be affected by higher structural deformation, which signifies higher groundwater infiltration potential zones and initiation of first-order drainage networks. This information is vital for water resources management in the sub-catchment.

**Keywords:** Lineament, DEM, Extraction, Azimuth-Altitude Ratio.



## Introduction

Lineaments are linear features that occur on the earth's surface, distinct from the surrounding features, and reflect what the unseen subsurface can be (O'Leary et al., 1976). They can be seen in natural forms such as faults, joints, and fractures in rocks which are all secondary porosity and act as both conduits and storage for groundwater, they could also be artificial features such as roads, tracks, and buildings (Alshayef et al., 2017). According to Alonso-Contes (2011), geological lineaments can have their origin in structural or geomorphologic activities and when captured by certain satellite sensors, they can be differentiated from anthropogenic effects such as roads, trails, and fence lines. They are expressed in the form of texture, color, and topographic changes. In a crystalline terrain, the presence of lineaments and their density is a major determinant of groundwater recharge and occurrence (Abdullateef et al., 2021).

Morphometric mapping and analysis of genetic (geologic and geomorphologic) lineaments have long been applied in a wide range of applications. Genetic lineaments include geologic (due to tectonic activities) and geomorphologic or topographic (due to geomorphic processes). Lineaments have wide implications and applications including lithological mapping, hydrothermal alteration zone mapping, morphotectonic and active fault detection, mineral prospecting and exploration, water resource exploration and abstraction, hydrogeomorphology, hazard assessment, geological mapping (Das et al., 2017; Das and Pardeshi 2018; Shandini et al., 2020).

The conventional methods of mapping lineament are usually done by monitoring the faults in the field. This makes it difficult to identify all existing lineaments in an area of interest (Kassou et al., 2012). Experts and researchers have shown the value of remote sensing data for lineament mapping and the identification of geological structures (Scanvic, 1983; Alshayef et al., 2017; Bouchra et al., 2019). Aerial photographs, satellite images, and in recent times digital elevation models have been

used for the extraction of lineament using different approaches (Szynkaruk et al., 2004).

Two methods are proposed for identifying and extracting lineaments from remote sensing data: manual and automatic. The manual method consists of enhancing the tonal differences, and then visually interpreting the lineament segments of the satellite image. This technique is efficient but it is relatively time-consuming, tedious, and more importantly highly subjective (Alshayef et al., 2017; Adaviruku & Oyatayo, 2023). The automatic method uses computer software and algorithms to detect and extract lineaments automatically. Although this technique is quick, it generates many insignificant lineaments and must go through a post-processing phase that includes validation and removing anthropic lineaments. As a result, some manual processing should be incorporated (Abdelouhed et al., 2022). The downside of the automatic lineament extraction technique is that it does not discriminate linear features, it treats roads, tracks, and water systems as lineaments with geologic structures.

Different satellite images are used for lineament extraction most especially Landsat images and LiDAR data (Albert & Taofeek, 2016). Image processing techniques such as Principal Component Analysis (PCA), Minimum Noise Fraction (MNF), and Band rationing, improve and enhance satellite images to facilitate the differentiation and characterization of structural geological elements for visually interpreting lineament based on identification in the satellite images (Mathew & Ariffin, 2018).

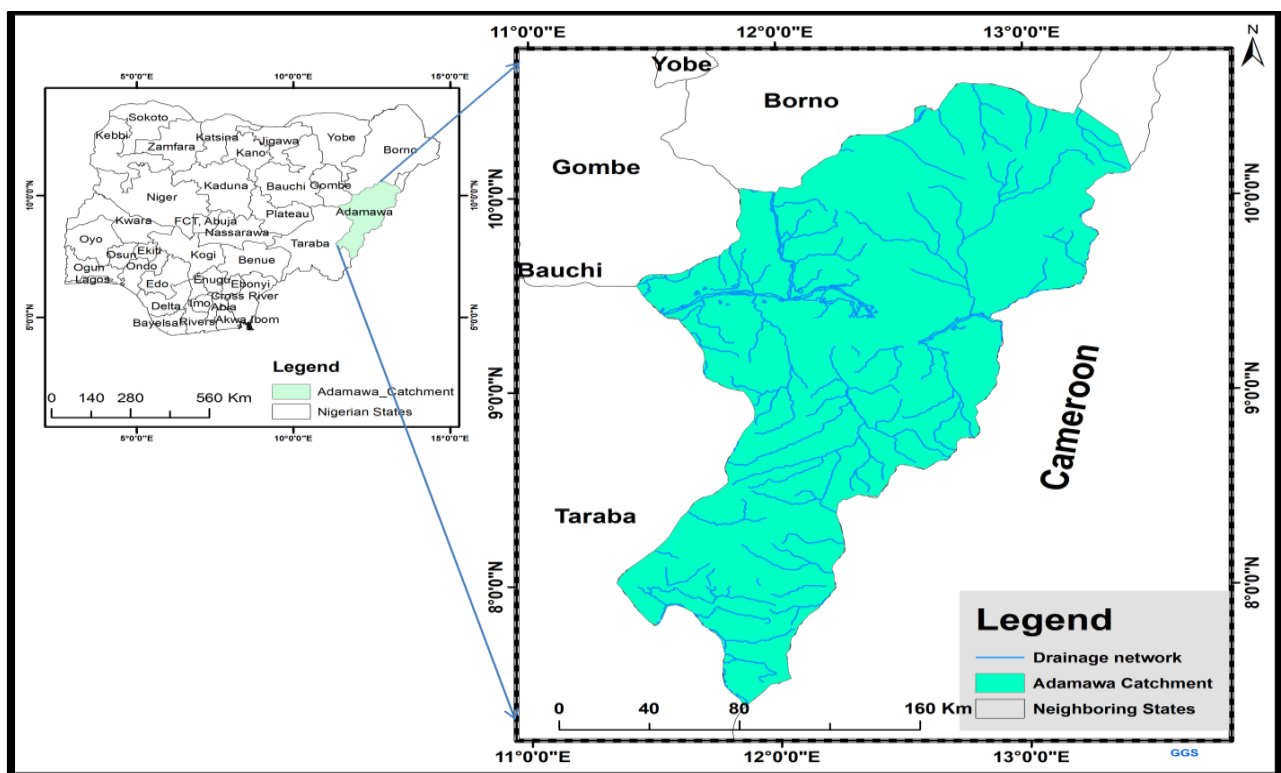
Considering the importance of lineaments, the present study is carried out to understand the spatial distribution, orientation, density, and influence of lineaments wide range of implications and applications in geology, hydrogeology, and geomorphology. The focus of this study was on structural or geological lineaments and their implication on hydrogeology and hydrogeomorphology in the Adamawa sub-catchment in the Upper Benue River Basin with a view to providing timely data to policy, decision-makers, and stakeholders. This can guide the process of making well-

informed decisions that can impact the availability and use of water resources in the study area.

### The study area

The Adamawa sub-catchment of the Upper Benue River Basin is located between Latitudes  $07^{\circ}20'00''$  and  $10^{\circ}30'00''$  North of the Equator and Longitudes  $11^{\circ}20'00''$  and  $13^{\circ}30'00''$  East of the Greenwich Meridian (Figure 1). It shares a boundary with Taraba State in the southwest, Gombe State in the northwest, Borno State to the North, and an international boundary with the Cameroon Republic along its eastern side. It has a land area of about  $38,741 \text{ km}^2$  (Adebayo, 1997). The study area is overlain by both

sedimentary and crystalline geological formations. Majorly, the basement rocks such as Older Granites, Migmatite, and Gneisses of Precambrian age are the oldest and overlain in some localities by sedimentary rocks of Upper Cretaceous to Quaternary (Carter et al., 1963). According to Ahmed et al. (2017), the crystalline rocks of Adamawa consist of highly metamorphosed rocks mainly the Migmatites, Gneisses, and Older Granites, and these rocks outcrop extensively in the study area occupying a greater percentage of the rocks in the State at large. River Gongola is the major tributary of river Benue that drains through the Adamawa sub-catchment of the Upper Benue river basin.



**Figure 1. Adamawa Sub-Catchment of the Upper Benue River Basin**

**Source:** Adapted from Upper Benue River Basin map

### Material and Method

The primary data used for this study was the Digital Elevation Model (DEM) of the study area acquired from USGS via Earthexplorer. Since the study area extends over several tiles of the images, eight tiles of the DEM were used by

mosaicking the eight tiles into a single image. Thereafter image sub-setting was carried out on the mosaicked image to extract out only the area of interest which is the Adamawa sub-catchment of the Upper Benue River Basin. Sub-setting was done in ArcGIS 10.8 (ArcMap) environment using the spatial analyst tool. The lineament map

for this study was extracted from the image of the area by producing the hill shade of the area to enhance the feature extraction at various Azimuth and Altitude ratios (315:45, 200:50, 150:65 and 100:60) in ArcGIS 10.8. This was followed by manual digitization of the lineaments. This approach is now popularly used by many spatial tutors but is yet to be documented (GIS and Remote Sensing Basics, 2020; Geojay GIS Solution, 2023)

Varying the Azimuth and Altitude ratio as done above enhanced the image for easier and better interpretation, and in addition, this technique helps the analyst to easily distinguish other linear features on the image that are not geological structures like man-made structures and even river channels (Figures 2 to 5). The line density function of the spatial analyst tools was then used to produce the lineament density map from the lineament. The Statistical Analysis of Lineaments was done on the lineaments network geometry to identify the dominant directions. Based on the work of Kassou *et al.* (2012), the statistical analysis of the lineaments was used to study the lineaments network geometry and to identify the dominant directions at the regional level. In this study, the identified lineaments were subjected to lineament statistical analysis in Rockworks17 software. This determined the

different structural lineament trend and structural orientations in the study area.

## Results and discussion

### Hill-shades at varying Azimuth and Altitude ratio

The essence of varying the Azimuth and Altitude ratio in producing the hill shade is to enhance the linear features of the image for better visualization and extraction. The result of this process as shown in Figures 2 to 5 revealed that at various Azimuth and Altitude ratios, the features on the image become more distinguished. It can be seen that at 315:45 Azimuth and Altitude ratio, the Hill-shade produced revealed the major lineaments more clearly compared to other ratios but minor lineaments were undermined. However, the 200:50 Azimuth and Altitude ratio produced an image more enhanced, revealing salient lineament that was not captured at the 315:45 ratio. At 150:65 Azimuth and Altitude ratio, the image was duller; however, some lineaments that were missing in the previous ratios could be detected. The final ratio used in this study was 100:60, at this Azimuth and Altitude ratio the image had the least distinguished lineaments.

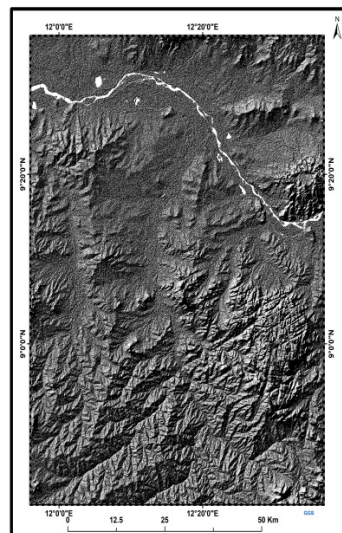


Figure 2. 315:45 Azimuth and Altitude ratio

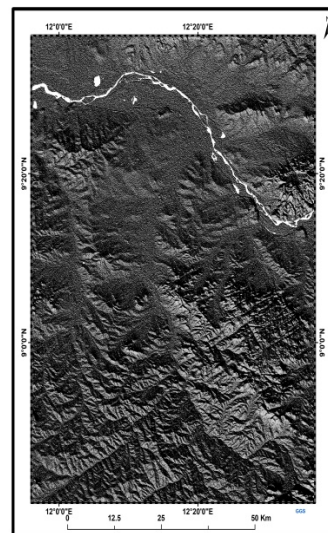
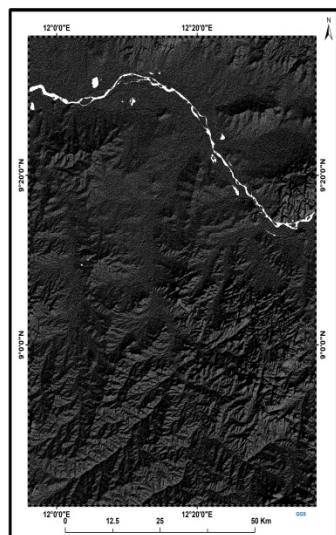
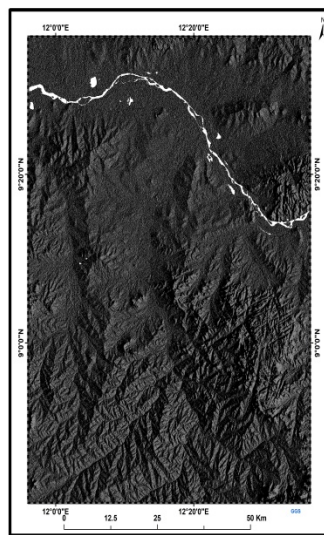


Figure 3. 200:50 Azimuth and Altitude ratio





**Figure 4. 150:65 Azimuth and Altitude ratio**



**Figure 5. 100:60 Azimuth and Altitude ratio**

### Lineament and Lineament Density

A total number of 343 lineaments were extracted from the Digital Elevation Model (DEM) with a total length of 2, 897.81km. The result of the lineament extracted is presented as a lineament map in Figure 6. It was observed that lineaments are more concentrated in the northern and southern parts of the study area while the central region has sparse lineament. This may be due to the fact that the central region of the Adamawa sub-catchment of the Upper Benue River Basin is overlain by sedimentary rocks of Upper Cretaceous to Quaternary (Carter et al., 1963). Similarly, the length of lineament in the study area ranges between 0.14km and 55.07km. The lineament length found in this study is contrary in range to the work of Chandramohan and Vignesh (2019) which showed that the lineament length of Palani Taluk, India varies from 0.11km to 3.25km. The longest lineament was found in the southern part of the Adamawa sub-catchment of the Upper Benue River Basin.

Lineament density also referred to as Bilineament length density (Greenbaum, 1985), is defined as the total length of lineaments per unit area. The density of lineaments provides valuable information regarding the intensity of tectonic deformation (Hung et al. 2002; Das et al. 2003), rock fracturing and shearing (Edet et al., 1998; Kanungo et al., 1995; Chandrashekhar

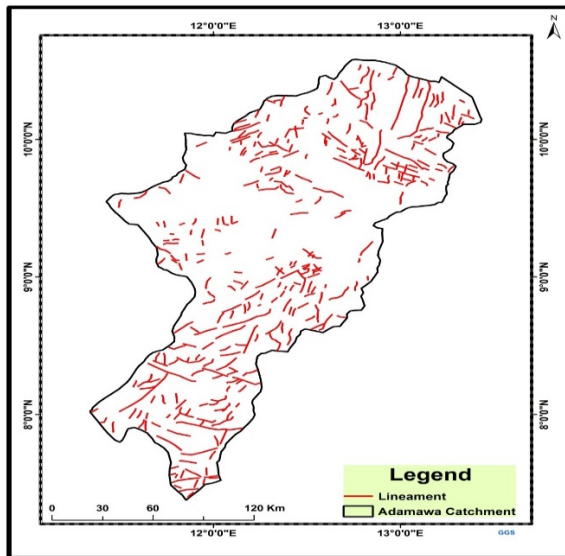
et al., 2011), permeability of rock (Masoud and Koike, 2011) and groundwater potential (Nag 2005; Das et al., 2017). Some studies (Djeffal et al., 2023) have established that geologic and topographic factors control the lineament density and orientation.

Figure 7 shows that the lineament density ranges between 1km/km<sup>2</sup> and 5km/km<sup>2</sup>. This implies that for every km<sup>2</sup> coverage in the study area, there is about 1km-5km lineament present. The denser lineament signified the intensity of rock fracturing, which is a prerequisite for development of conduits for groundwater in basement complex. As noted from the result of lineament density, the northern and southern parts of the study area have more lineament density which implies for instance that if one is interested in groundwater potential, these areas may likely have more potentiality than others, other factors being equal.

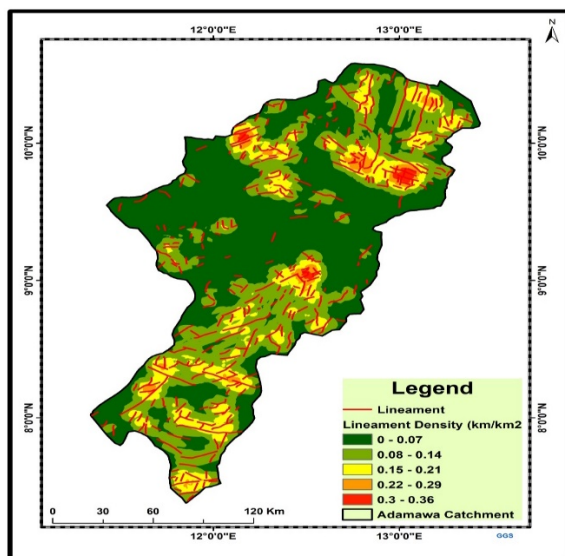
### Lineament Statistics and Trend

The rose (azimuth-frequency) diagram of the lineaments drawn from the extracted lineaments shows the following trends: N-S, NNE-SSW, NE-SW, E-W, SE-NW, and SSE-NNW. The NNE-SSW lineaments direction is the dominant trend in the study (Figure 8). This contradicts the work of Amigun et al. (2015) who found that the lineaments trend in Okene and its environs

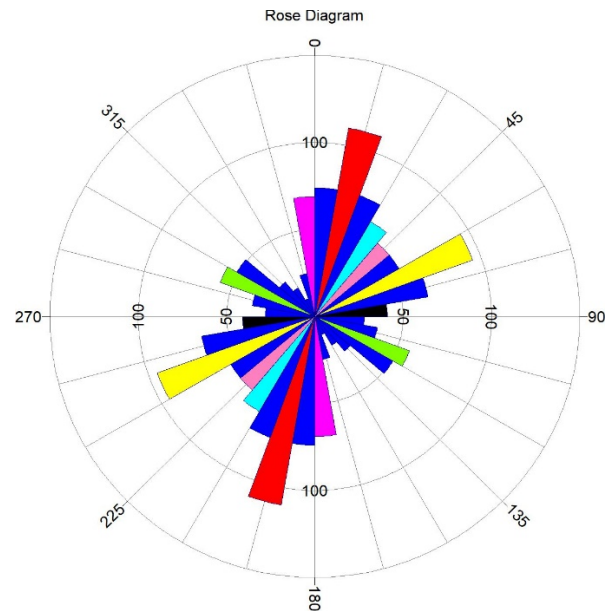
were in the NE-SW and NNE-SSW directions. This deviation may likely be due to the fact that they used few lineaments to generalize their study area (Kogi central), whereas the current study took enough samples for the entire Adamawa sub-catchment of the Upper Benue River Basin. The result of this lineament trend is similar to the findings of Coulibaly *et al.* (2021) who found that the trend of lineament in Béré region were in the direction of N-S, NE-SW, E-W and NW-SE direction.



**Figure 6. Lineament of Adamawa sub-catchment of the Upper Benue River Basin**



**Figure 7. Lineament Density of the Study Area**



**Figure 8. Rose Diagram of Lineament Orientations in the Study Area**

## Conclusion

The analysis of the extracted lineaments from four (4) different azimuth angles – altitude ratios (315:45, 200:50, 150:65 and 100:60) indicates that extraction of lineament is more accurate and comprehensive if different azimuth angles are considered instead of a particular azimuth angle. The map showing lineament density zones can be implied for groundwater potential zones. The areas of high lineament density might have greater groundwater infiltration potential but should be avoided for engineering constructions due to high vulnerability of erosion and slope failure.

## Recommendation

In order to give room for validation and for further studies, this study therefore recommends that conventional and any other methods should be used for lineament mapping of the Adamawa sub-catchment of the Upper Benue River Basin. It is also recommended that for groundwater development in the area the result of this study should be incorporated with other factors for efficient groundwater exploitation.

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